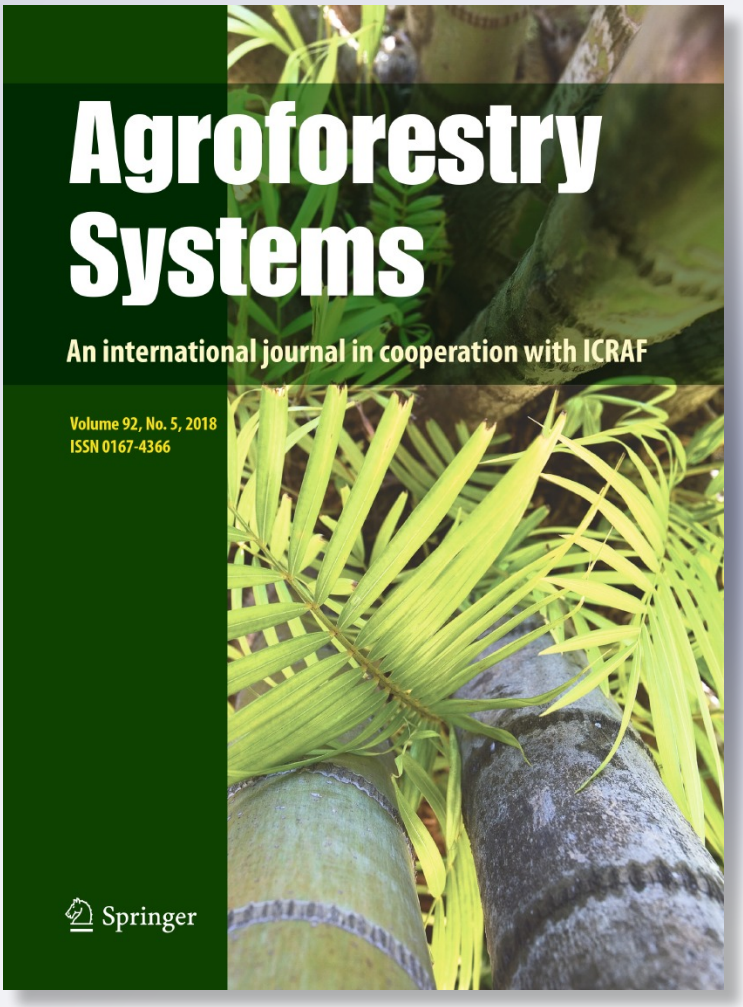
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Wood biomass recovery from chestnut orchards: results from a case study

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Abstract Sweet chestnut has been for many centuries essential to human diet in large areas of Southern Europe. Its cultivation was abandoned in the last century, but is at present under restoration for socioeconomic reasons, representing also an opportunity for allocating woody residues to the energy sector. A little is known from the literature about sweet chestnut pruning, and the aim of the study was to assess the biomass yield and quality, the productivity and costs of the system as well as its energetic balance between inputs and outputs. The yield of recovered wood material amounted to between 22.3 tonnes of dry matter per hectare (t_{dw} ha^{-1}) and 33.3 t_{dw} ha^{-1}. Time consumption for pruning has been related to trees' DBH, so detecting a linear relationship, although weak, between tree size and the time spent for maintenance. Productivity expressed as t_{dw} per hour varied according to the site and the operating systems adopted accordingly. Costs for the whole chain, excluding transport to the plant facility amounted to 113 or to 430 € t_{dw}^{-1} depending on the terrain relief and the presence of an underbrush to be cleaned. Wood chips distribution in size classes provided a material unsuitable for non-industrial due to the percentage of oversize particles, probably due to the high presence of twigs. The energy ratio resulted of 30:1 and 21:1 for the two sites. Transportation had the main impact in terms of energy, followed by extraction and chipping phases.

Keywords Chestnut orchards · Energy balance · Pruning · Recovery systems · Wood biomass

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Introduction

The reforms of the Common Agricultural Policy (CAP) have introduced a financial instrument to strengthen the EU's rural policy through the European Agricultural Fund for Rural Development (EAFRD). This instrument, which was established by Regulation (EC) 1290/2005, aims at improving the competitiveness of agriculture and forestry, the environment and the countryside, the quality of life and the management of economic activity in rural areas (Cofas and Toma 2014). Member States have been called to draw up a national strategy plan in line with the strategic guidelines adopted by the Community according to their own thematic and territorial priorities. The rural

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development programme should identify the needs of the area covered and describe a coherent strategy to meet them, taking into account the priorities of EU rural development (Sandu 2014). In particular, Axis 4 (LEADER) contains a package of measures related to the implementation of local development strategies, through public–private partnerships called "local action groups". As other Member States, Italy has selected, at Regional scale, a number of agricultural supply chain to be sustained as cereal growing, animal husbandry, olive oil and wine growing or nursery gardening.

An important supply chain within the Measure 124, Axis 4—Regional Development Programme of the Tuscany Region (Italy)—affects the sweet chestnut groves recovery, needed after the abandonment of orchards as a result of species diseases (Maltoni et al. 2012) or depopulation of rural areas occurred at a European level started after the Second World War (Álvarez-Álvarez et al. 2010). Chestnut orchards belong to the European traditional landscape, represent a protected habitat according to the Directive 92/43/EEC (Pezzi et al. 2011), and a possible reduction of their extension may indicate a forest degradation occurring as a result of the natural succession dynamics or the invasion by alien species (Díaz-Varela et al. 2011). Italian chestnut stands cover 10% of the national forest surface, and Italy is the second producer for nuts in the EU and fifth at international level, with 52,000 tonnes (t) of fruits yearly harvested. Leader in the productivity international rank is China with 1 650 000 t, followed by the Republic of Korea (70,000 t), Turkey (59,789 t) and Bolivia (5700 t) (FAOSTAT 2012). In Tuscany, chestnut (*Castanea sativa* Miller) cultivation that has dominated the rural economy for centuries (Agnoletti 2007) since medieval times (Conedera et al. 2004) is nowadays spread on almost 34,000 hectares (ha) against the 147,568 ha of the whole Italian territory occupied by this species (INFC 2005). Regular pruning should occur every 3 or 5 years (Bariselli et al. 2013), but despite former measures had supported chestnut tending (Fratini and Riccioli 2009), often the shortage of financial resources leads to postpone any maintenance for several years. As a result, a remarkable amount of wood biomass has to be handled between one tending operation and the following. Traditionally, branches with a diameter bigger than 5 cm are severed in pieces for firewood purpose, although chestnut wood is considered of a little market value. Smaller material is usually burnt on the premises as it happens for other woody agricultural wastes (Recchia et al. 2009; Spinelli et al. 2010, 2014). However, biomass open burning entails several drawbacks (Estrellan and Iino 2010). The combustion of pruning residues releases pollutant substances (Gonçalves et al. 2011) in a remarkable amount at local level, mostly due to the presence of polycyclic aromatic hydrocarbons, polybrominated dibenzo-p-dioxins and dibenzofurans (McIlveen-Wright et al. 2001), and solid particles (Keshtkar and Ashbaugh 2007) even if lower compared to car traffic (Darley et al. 1966).

In the project BIOCASPO, funded within the LEADER Measure, it was requested to transfer the technical knowledge of the scientific partner to recover the chestnut pruning for energetic use. Very little it has been investigated in the literature about the biomass produced during pruning operations in chestnut stands so far. The Authors' aims are (1) to assess the biomass yield in chestnut orchards and its quality for feeding non-industrial facilities; (2) to calculate productivity and costs of the system; (3) to estimate the biomass yield by means of a visual assessment and determine any possible relationship between operating time and the characteristics of the orchards; (4) to present an energetic balance between inputs and outputs embedded into the system.

Materials and methods

Study sites

Three chestnut groves of 2834, 2423 and 2038 m^2 were selected in the Amiata area, Tuscany region, close to each other but different mostly from terrain slope and tree density. Age of trees varied considerably among sites and even into the same plot, as usual chestnut trees are replaced when unproductive. In the same property, it is not unusual to find young subjects beside adults. For this reason, age was not taken into consideration as a parameter in the present study. Tests were conducted on one plot per grove, and namely: site A (N42°53′28″ E11°29′01″), site B (N42°53′52″ E11°28′51″) and site C (N42°51′42″ E11°31′34″), whose characteristics are shown in Table 1.

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Table 1 Description of stand characteristics in the different sites: A, B and C

Site	A	B	C
Area (m^2)	2834	2423	2038
Average ground slope (min–max) (%)	10.3 (9–13)	43.3 (31–60)	27.0 (21–36)
Density (trees ha^{-1})	120	214	157
DBH (cm)	46.3	35.4	56.0
Height (m)	9.4	5.4	11.5
Yield (t_{dw} ha^{-1})	33.3	22.3	25.5

DBH diameter at breast height (1.3 m), t_{dw} tonnes dry weight

While in sites A and B (Fig. 1) the whole pruning and biomass recovery chain was studied, this was not possible in site C, which was used to provide information, besides stand characteristics, about the pruning and chipping stages, besides the drying time of wood material. The study was conducted in 2014—from January to May.

Tree pruning and gathering of wood biomass

In each plot, from 30 to 60 chestnut trees were pruned by professional tree-climbers who trimmed one tree at a time using ladders, ropes, climbing harness and small-sized chainsaws. In site A, the branches fallen on the ground were cut to a length of about 3.5 m and loaded on the trailer of a farm tractor equipped with a crane and a grapple. Wood material was then forwarded, downloaded and stacked in piles at an intermediate landing on a secondary road. Here, the piles were picked up by a truck equipped with a crane and a log grapple, then moved to a chipping centre, 7.4 km away. Chipping was carried out by an industrial drum chipper driven by a powerful tractor and fed by a wheeled loader equipped with a crane and a log grapple. Wood chips were directly blown in the bin of a truck.

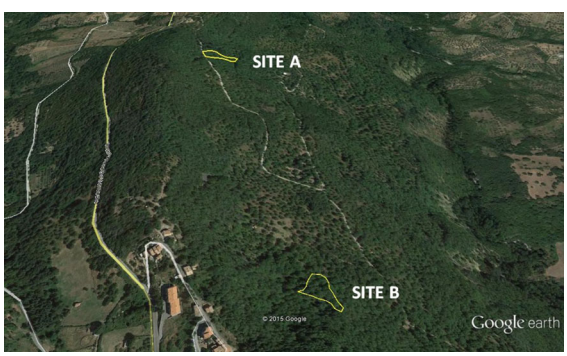


Fig. 1 Two out of three chestnut orchards under study

Site B was characterized by a thick underbrush formed by fern, heather and bramble that was cleared before pruning. This stage was realized by a team of two people of whom the first one used a brushcutter to eliminate shrubs, and the second one helped by moving and stacking the material in heaps. Once the undergrowth was cleared, the first operator, equipped with a medium size chainsaw, cut the wild chestnut trees of no interest for the orchard and left them on the ground for being extracted with the pruning. Differently from the undergrowth material, wild chestnut trees were recovered as wood for energy. Due to the slope, pruning was extracted with a winch mounted on a farm tractor, and then pulled uphill or downhill according to the distance of the forest track, accessible for transport means. The use of a farm tractor and a winch is a quite common technique for wood extraction in Italy (Picchio et al. 2009; Magagnotti et al. 2011), because it combines inexpensiveness and ease of use. Once the branches had reached the road, they were loaded on the truck and transported to the chipping centre for comminution.

In site C, pruning was performed as in the other two sites, but extraction was carried out by hand. Transportation to the comminution centre and chipping stage was operated as already described for sites A and B, but only the pruning and the comminution phases were recorded.

Productivity

Trees diameter was measured by means of a caliper, trees height by means of the Haga hypsometer. The study area—containing the experimental plots—was measured with a navigational hand-held GPS device, the distances for extraction and forwarding phases

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were measured by means of a hip chain, the odometer installed on tractors or the GPS.

Cut branches were grouped in four classes of diameter: up to 5 cm (D1), up to10 cm (D2), up to 15 cm (D3), bigger than 15 cm (D4); and in four groups of length: up to 1 m (L1), up to 2 m (L2), up to 3 m (L3), longer than 3 m (L4). In order to estimate the biomass available from every single tree, the Authors used the frequencies of diameter and length recorded in each class to calculate the related volume, and finally the mass, by applying an appropriate value of mass density (Wood Fuel Handbook 2008). For each branch, the central value for the group of reference (e.g. 2.5 cm for the first set of diameters, 0.5 m for the first set of lengths) was taken, multiplied by the number of elements contained in each class and divided by the total number of elements. In this way an average volume for each branch of every pruned tree was estimated, which was considered as a replicate. The amount of wood biomass measured at the end of the recovery chain would be used as a "retrofit" of the estimation itself. Since the match between single diameters and lengths was unknown, the most "pessimistic" and the most "optimistic" configurations were the following: Configuration 1 (D1, L4) (D2, L3) (D3, L2) (D4, L1); Configuration 2 (D1, L1) (D2, L2) (D3, L3) (D4, L4). Of course any other combination of the factors was possible, and the tree volume would be contained within the two configurations bounds.

Working times were taken with a stopwatch and carried out at elemental level (Magagnotti and Spinelli 2012) to evaluate the system productivity and to identify those variables that are most likely to affect it. Field productivity was measured as Gross Productive Machine Hours (PMH_{15}), including all delays up to 15 min, and Net Productive Machine Hours (PMH_0), excluding delays. This was done to avoid the baffling nature of delay time, which is typically irregular (Spinelli and Viesser 2009) if based on a short-term observations.

Wood chips loads were taken to a public weighbridge to measure the mass produced.

Biomass quality

Three samples of wood chips, 1000 g each, were taken from the truck bins filled up at each experimental plot to determine the material's moisture content (m. c.) and particle size distribution. The determination of m. c. was conducted on subsamples, collected in sealed bags and weighed fresh and after drying at 105 °C to constant mass, according to EN 14774-2:2010. Moreover, wood chip quality was assessed by sieving the wood chips according to the EN 15149-1:2011 standard. Sieves were used to separate chips in the following length classes: >63 mm, ≤63–45 mm, ≤45–16 mm, ≤ 16–8 mm, ≤8–3.15 mm, <3.15 mm. Chips longer than 100 mm were measured separately. Each fraction was then weighed with a precision scale.

Cost calculations

Machine costs were calculated with the method described by Miyata (1980), on an estimated service life of 2 years for the chainsaws and the brushcutter, 10 years for all the machines except the truck (5 years). At the end of their service life, all machines were considered to retain 20% of the original value except for the chainsaws and the brushcutter. Labour cost was set to 18 € per scheduled machine hour, including indirect salary costs. Interest rate was set to 8% for chainsaws and brushcutter, and 4% for all the other machines. The costs of fuel, insurance, repair and service were obtained directly from the operators. The calculated operational costs of the tractor, the truck, the loader and the chipper were increased by 20% in order to include relocation, overheads and administration costs (Hartsough 2003).

Energy calculations (sites A and B)

As stated by IFIAS (1975) energy analysis concerns the discrepancy between the change in thermodynamic potential in a specific process and the "embodied energy" requirements calculated by the accounting procedure. The GER (Gross Energy Requirement) method takes into account the fossil energy flows contained in a system, both direct and indirect (Biondi et al. 1989). Direct energy concerns the energy employed during the production process; indirect energy is the energy embedded in machines and tools (Picchio et al. 2009) deployed in the production of goods. The energetic efficiency of the system is the ratio between inputs (direct and indirect) and outputs, represented here by wood biomass.

In the present study, only the use of non-renewable energy has been considered. Renewable

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energy in form of solar radiation which is the basis for the build of tree biomass was not accounted for in the balance.

Direct energetic inputs, that is fossil fuel and lubricant consumption, were measured for the whole recovery chain. In the pruning stage, every working session started with a full chainsaw tank and ended when it was empty. At that stage, the chainsaw was weighed with a portable lab scale, filled with mixed fuel and weighed, then filled with oil, and weighed again. This procedure was applied in order to separate the fuel from the lubricant. For the other machines, the fuel consumption was measured by starting each working day with a full tank and refilling the tank at the end of the day. Energetic values for mixed fuel, lubricant and diesel (Biondi et al. 1989; Guadagni 2010) were multiplied by the amount consumed into the system. Lubricants and greases for machines were calculated as a fraction of the fuel, assuming 3 kg every 100 kg of diesel for tractors, lorries and trucks (Biondi et al. 1989).

Indirect energetic inputs were calculated by multiplying the energetic value embedded in tools and machines (Biondi et al. 1989) deployed in the recovery chain (MJ), by the their mass (kg), dividing by total service life (h) and finally multiplying by the use (h) in the trial (Volpi 1992).

In accord with the established conventions of industrial energy analysis, energy inputs associated with human labour were ignored (Matthews 2001).

Energetic output was obtained multiplying the dry wood biomass by the gross caloric value of chestnut wood (Telmo and Lousada 2011).

Statistical analyses

Data were analysed with the free statistical software R (version 3.1.2), using the open source Rstudio integrated development environment (version 0.98) under Mac OS X Mountain Lion. An explorative analysis was conducted with the aim of determining the existence of correlations between variables, both primary (directly measured, like tree diameter and length) and secondary (as the estimated length and diameter of the cut branches). Afterwards, the variance structure of relevant variables, such as the productivity of the pruning phase, was investigated by means of parametric and non-parametric methods, as described in the following section.

Results

Productivity

The yield of recovered wood material amounted to 50.5 fresh tonnes (t_f) per hectare (ha^{-1}) at 34.1% of m. c. or 33.3 dry weight tonnes (t_{dw}) ha^{-1} in site A, to 35.5 t_f ha^{-1} at 37.2% of m. c. (22.3 t_{dw} ha^{-1}) in site B, and 33.1 t_f ha^{-1} at 23% of m. c. (25.5 t_{dw} ha^{-1}) in site C.

In order to analyze statistically the yield of wood, it would be needed to know the single mass obtained from every tree, but the only information available was the total mass produced per site, the number of pieces in classes of diameter and length and the DBH for every tree.

The aim was to estimate the pruned mass per single tree through a model based on the number of elements, gathered in classes of diameter and length and using the measured weights per each site as a retrofit.

Since there was no match between a specific diameter and its relative length, there were two extreme possible configurations, where any combination was contained in:

Configuration 1: (D1, L4) (D2, L3) (D3, L2) (D4, L1)
Configuration 2: (D1, L1) (D2, L2) (D3, L3) (D4, L4)

By allocating average values a_1, a_2, a_3, a_4 to the diameter classes and average values b_1, b_2, b_3, b_4 to the length classes, pertaining volumes were as follows:

$$V1 = \pi a_1^2 b_4/4 + \pi a_2^2 b_3/4 + \pi a_3^2 b_2/4 + \pi a_4^2 b_1/4$$

$$V2 = \pi a_1^2 b_1/4 + \pi a_2^2 b_2/4 + \pi a_3^2 b_3/4 + \pi a_4^2 b_4/4$$

Obviously, any other combination of couples (D_i, L_j) was possible, and the "real" volume would be contained within these bounds. In general, we had:

$$V_m = \frac{A_{\text{tot}} B_{\text{tot}}}{N},$$

where V was the average volume, A the average area, B the average length and N the elements of the sample.

The volume was then multiplied by the standard mass density for chestnut, and finally converted in dry weight. The estimated amount of biomass turned out to be noticeably lower when compared to the measured one. So the estimated values were multiplied

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by a coefficient resulting from the ratio between the two amounts. Data resulted to be moderately heteroscedastic and residuals appeared not normal, so the values were transformed according to Box and Cox (1964) method, in order to normalize the residuals.

The pruning time, that is the cutting time plus the displacement of the operator on the tree, has been related to trees' DBH, so as to detect possible interaction between the tree dimensions and the time spent for maintenance.

Figure 2 shows that such an interaction exists in an average sense the pruning time linearly depends on the tree size. The linear relationship between the pruning time and DBH is rather weak (the explained variance is about 30%) but nonetheless statistically meaningful.

The results for gross and net hourly productivity are reported in Table 2. Site A emerged more productive as regards pruning and transport operations compared to site B, while site B and C were very close to each other concerning pruning but very different about chipping. Regarding the latter, site A and site C had both a production above 10 tonnes produced per hour (dw) (Tables 3, 4).

The productivity of the whole pruning phase (t_{dw} PMH_{15}^{-1})—that is climbing, cutting, displacement on the tree, descending, collecting the equipment, moving to the next tree—was analysed in the three sites. The box plot in Fig. 3 compares the productivity of the three sites.

At a first look, the productivity of the "A" site appears larger than the other two, which have a similar

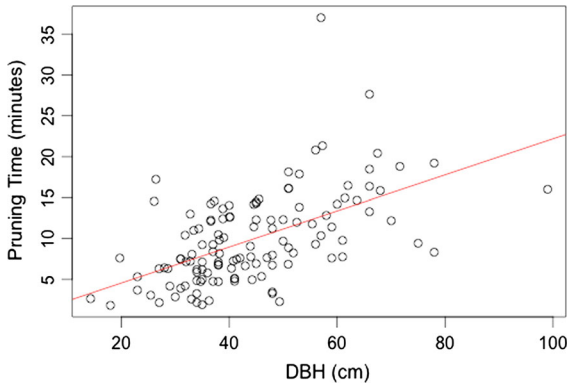


Fig. 2 Relationship between pruning time and trees' DBH (diameter at breast height) Adjusted R-squared: 0.3011 Equation: Pruning time = a*DBH + b with: $a = 0.22035$ $b = 0.15306$

median value. Neglecting the outliers, the data dispersion appears almost homogeneous through the sites, suggesting a moderate heteroscedasticity.

The yield of wood biomass was compared within the three sites in order to detect any statistical difference. Also tolerating the slight difference among the variances of the three sites (the Bartlett's test tells they are heteroscedastic, while the Levene's one does not allow to discard the null hypothesis, suggesting that data are border-line), the problem with data is their deviation from normality. This last problem can be solved by transforming the variable productivity with the procedure developed by Box and Cox (1964) and Venables and Ripley (2002). The transformed productivity, given by the expression:

$$p_T = \frac{(t_{dw} PMH_{15}^{-1})^{\lambda} - 1}{\lambda}$$

with $\lambda = 0.263$, has normal residues but it is still moderately heteroscedastic.

Figure 4 shows the results of the ANOVA on the transformed productivity, using the site as a factor. In terms of the Tukey Honest Significant Difference, the production of the "A" clearly differs from those of sites "B" and "C", confirming the first impression based on the analysis of the box plot.

A more rigorous, although less powerful, non-parametric test based on the medians of the original, non-transformed, data (Kruskal and Wallis 1952) confirms the above result. The Kruskal–Wallis (KW) test applied to the gross productivity as a function of the site reveals a statistically significant difference among the productions in the three sites ($p = 0.0012$), as one can reasonably expect from the analysis of the box plot shown in Fig. 3. As the KW test is significant, a post hoc test on group pairs is mandatory to verify if the feeling that site "A" results differ from those of sites "B" and "C" is supported by the statistical analysis. By performing three pairwise Wilcoxon–Mann–Whitney tests (Siegel and Castellan 1988), with Bonferroni correction for multiple comparisons, the null hypothesis (concerning the equality among the mean ranks of the three sites) can be rejected at a level 0.05 only for the comparison between sites "A" and "B" (adjusted $p = 0.0007$) and between sites "A" and "C" (adjusted $p = 0.0054$), while it cannot be rejected for the remaining comparison (adjusted $p = 1$). This result is confirmed using the Dunn's test (Dunn 1964), as the following table shows.

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Table 2 Gross and net productivity in the three sites under study expressed as tonnes of dry weight per hour

Stages	Gross and net hourly productivity					
	SITE A		SITE B		SITE C	
	t_{dw} PMH_{15}^{-1}	t_{dw} PMH_{0}^{-1}	t_{dw} PMH_{15}^{-1}	t_{dw} PMH_{0}^{-1}	t_{dw} PMH_{15}^{-1}	t_{dw} PMH_{0}^{-1}
Pruning	1.0	1.1	0.4	0.5	0.6	0.6
Cleaning	–	–	0.2	0.2	–	
Processing	1.5	1.5	–	–	–	
Extraction	1.1	1.3	1.0	1.2	–	
Transport	2.5	2.5	1.6	1.6	–	
Chipping	14.1	14.1	9.1	9.1	19.7	19.7

Table 3 Dunn's test on gross productivity in site A and site B

Col mean—Row mean	Comparison of gross productivity by site	
	A	B
B	−3.509044	
	0.0007	
C	−2.912031	0.257209
	0.0054	1.0000

Quality

The different moisture content recorded in the wood biomass was due to the time elapsed between pruning and chipping operations. The drying trend in wood material after 41, 105, and 188 days of storage is shown in Fig. 5.

Wood chips size distribution is reported in Table 4. Wood chips were analysed for allocation in commercial categories according to the UNI EN 14961-4:2011, but only one sample had the requested features for being assigned to P16B class. The others missed the specifications as woodchips for non-industrial use both in terms of main and coarse fractions.

Costs

Costs were evaluated for sites A and B only, and they were quite different in the two orchards. Site B when compared to Site A resulted more expensive in all of the processes carried out: +56% for pruning, +3% for extraction, +50% for hauling and +34% for chipping. Costs for pruning amounted to 28.0 € t_{dw}^{-1} in site A (one operator) and to 64.0 € $t_{dw}{}^{-1}$ in site B (two operators). The cleaning of the undergrowth was needed only in site B, and it costed 241.1 € t_{dw}^{-1}, accounting for the brushcutter and the piling of the material. In site A, since longer branches were more frequent compared to site B, a processing step was needed in order to reduce limbs in two or more pieces, to ease the loading of the material on the trailer and this stage costed 9.5 € t_{dw}^{-1}. The extraction, transport and chipping activities were present in both sites, however at different costs: 30.5 and 41.2 € t_{dw}^{-1} the

Table 4 Particle size distribution of wood chips according to UNI EN 15149-1:2011

Areas containing values within the range are left normal font, slightly out of the range are italics, out of range are bold

	Standard Class:P16B			
	Coarse (>45 mm) ≤3 (%)	Regular (16–3.15 mm) min 75%w (%)	Fines (<3.15 mm) ≤12 (%)	
	7.70	*73.70*	9.10	Site A
	2.20	*73.00*	7.60	Site A
	13.70	**68.80**	5.20	Site A
	2.90	77.73	5.50	Site B
	7.90	**52.80**	1.40	Site B
	10.00	**64.20**	1.80	Site B
Average	7.40	68.37	5.10	

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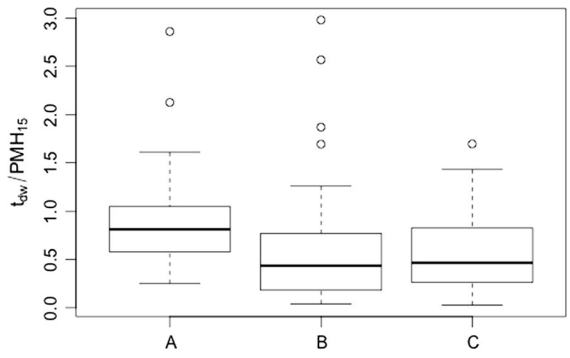


Fig. 3 Productivity of the pruning phase (t_{dw} PMH_{15}^{-1}) for the three sites

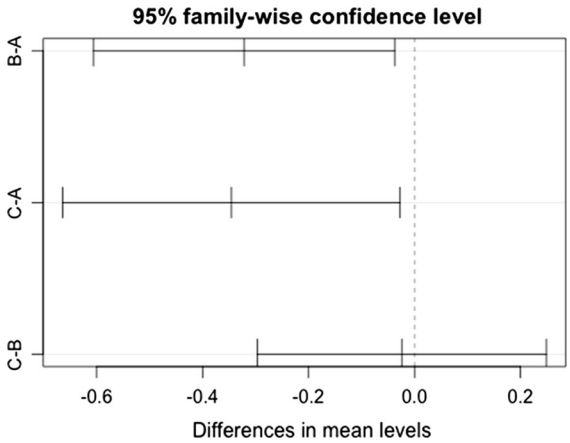


Fig. 4 ANOVA analysis on the transformed productivity versus sites

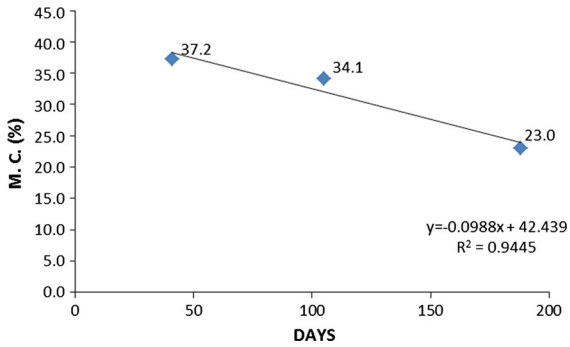


Fig. 5 Decrease of moisture content in wood material through natural drying

extraction, respectively, for site A and B; 29.4 and 59.4 € t_{dw}^{-1} the hauling of un-chipped material; 15.5 and 24.0 € t_{dw}^{-1} the chipping stage. The total cost per unit of surface amounted to 3757.2 € ha^{-1} for site A and 9569.2 € ha^{-1} for site B.

Energetic balance

Direct energy inputs for site A and site B have been calculated and shown in Table 5 starting from the amounts of fossil products used for any specific equipment and their gross calorific value.

In Table 6 the indirect energy inputs are presented for the two sites: mass (kg) and energetic value for each piece of equipment (MJ), expected life (h) and time spent on the trial (h).

In Table 7 the energy inputs, both direct and indirect, are reported for the recovering phases carried out in the two sites under study expressed as GJ per mass unit or as GJ per surface unit.

Direct energy inputs participate for 0.34 GJ t_{dw}^{-1} (9.89 GJ ha^{-1}) to the whole recovery operation in site A, and for 0.53 GJ t_{dw}^{-1} (13.63 GJ ha^{-1}) in site B. Indirect energy inputs accounted to 0.27 GJ t_{dw}^{-1} (7.92 GJ ha^{-1}) for site A and to 0.38 GJ t_{dw}^{-1} (9.67 GJ ha^{-1}) for site B.

As shown in Fig. 6, chipping in both sites employs most of the direct energy spent into the system (43.8 and 45.3%, respectively, for site A and site B), followed by extraction (41.2 and 42.5%).

Transportation has the main impact in terms of indirect energy, with 69.9% in site A and 79.4% in site B, followed by extraction (22.7 and 11.9%, respectively) and chipping (7.4 and 8.4%, respectively) phases.

Pruning and transportation require on the contrary very little direct energy (respectively 8.5 and 6.5% for site A, and 6.3 and 5.7% for site B).

The recovery and transformation of 1 t_{dw} of chestnut pruning chips require about 0.62 GJ in site A and 0.91 GJ in site B of non-renewable energy, mainly diesel fuel. As 1 t_{dw} of wood embodies 18.75 GJ of energy, this leads to an energy ratio respectively of 30:1 and 21:1 for the two sites.

Discussion and conclusions

Due to environmental constraints in mountainous areas, sweet chestnut has for many centuries been essential to human diet (Díaz-Varela et al. 2011). The present revival of its cultivation is driven by socioeconomic reasons but it represents also an opportunity for increasing the biodiversity, which is higher in groves than in coppice stands (Gondard et al. 2006) and allocating woody residues to the energy sector.

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Table 5 Energy direct inputs affecting each piece of equipment in site A and site B

Equipment	Fossil product	Gross calorific value (MJ/kg)	Fossil matter (Kg)	Gross energy content (MJ)
Site A Direct inputs				
Chainsaw	Mixed fuel	55.3	4.30	237.79
	Lubricant	83.7	1.90	159.03
Tractor & trailer	Diesel	51.5	18.48	951.72
	Lubricant	83.7	0.55	46.40
Truck	Diesel	51.5	3.40	173.04
	Lubricant	83.7	0.10	8.44
Tractor & chipper	Diesel	51.5	21.3	1097.17
	Lubricant	83.7	0.64	53.49
Loader	Diesel	51.5	1.40	73.14
	Lubricant	83.7	0.04	3.57
Total				2803.79
Site B Direct inputs				
Chainsaw	Mixed fuel	55.3	3.24	178.91
	Lubricant	83.7	1.42	119.25
Brushcutter	Mixed fuel	55.3	0.16	9.11
Tractor & winch	Diesel	51.5	25.20	1297.8
	Lubricant	83.7	0.76	63.28
Truck	Diesel	51.5	3.40	173.04
	Lubricant	83.7	0.10	8.44
Tractor & chipper	Diesel	51.5	25.2	1297.8
	Lubricant	83.7	0.76	63.28
Loader	Diesel	51.5	1.68	86.52
	Lubricant	83.7	0.05	4.22
Total				3301.64

There are not published studies based on chestnut pruning that can be directly used as a benchmark for our findings. The amount of pruning residues in the three sites showed significant variation, probably caused by differences in tree age, density and pruning intensity. The residual biomass resulted on average of 0.16 t_{dw} $tree^{-1}$ $year^{-1}$, assuming a ten-year time span between one pruning and the following. The overall amount corresponds to 2.7 t_{dw} ha^{-1} $year^{-1}$, a higher amount compared to almonds residues, yielding 1.34 t_{dw} ha^{-1} $year^{-1}$ (Velázquez-Martí et al. 2011), to olive pruning, producing from 0.3 to 2.5 t_{dw} ha^{-1} $year^{-1}$, (Spinelli and Picchi 2010; Spinelli et al. 2011a), to apple pruning, yielding 1.03 t_{dw} ha^{-1} $year^{-1}$ (Boschiero et al. 2016) or even to vineyard residues (Spinelli et al. 2012) whose production resulted being from 0.59 to 1.65 t_{dw} ha^{-1} $year^{-1}$. Yield appears to be linked to the specific conditions of the orchard, since sites A and C were trimmed by the same pruner and had also comparable dendrometric characteristics, but different amounts of biomass per unit of surface.

Productivity in the pruning phase cannot be compared to any other forestry practice, or because trimming is carried out with tools different from chainsaw, as hand pruners, handsaws, or pole pruners in high quality hardwood stands (Ryder and Moore 2013; Waring and O'Hara K 2005), or because affects coniferous plantations as described for example by Beinhofer (2010), or Erikkson (2004). Although an interaction exists between DBH and pruning time, this is too weak to allow the owner to forecast the time needed for maintenance. Maybe other parameters should have been taken into account, as vegetative

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Table 6 Energy indirect inputs concerning each piece of equipment in site A and site B

Equipment	GEV (MJ/kg)	Mass (kg)	GEV (MJ)	Life (h)	Trial (h)	Total GEV (MJ)
Site A Indirect inputs						
Chainsaw 1	92	3.0	276.0	1000	4	1.10
Chainsaw 2	92	4.6	423.2	1000	0.33	0.14
Chainsaw 3	92	4.0	368.0	1000	0.44	0.16
Tractor	92	4300	395,600	10,000	5.6	221.54
Grapple	69	1450	100,050	10,000	5.6	56.03
Trailer	69	60,00	414,000	10,000	5.6	231.84
Tractor	92	12,000	1,104,000	10,000	0.58	64.03
Loader	69	12,000	828,000	10,000	0.58	48.02
Chipper	92	10,100	929,200	10,000	0.58	53.89
Truck	92	26,000	2,392,000	5000	3.28	1568.62
Total						2245.38
Site B Indirect inputs						
Chainsaw 1	92	4.6	423.2	1000	2.8	1.18
Chainsaw 2	92	4.3	395.6	1000	3.7	1.48
Brushcutter	92	7.5	690.0	1000	3.7	2.55
Tractor	92	5600	515,200	10,000	5.26	271.00
Winch	69	250	17,250	10,000	5.26	9.07
Tractor	92	12,000	1,104,000	10,000	0.69	76.18
Loader	69	12,000	828,000	10,000	0.69	57.13
Chipper	92	10,100	929,200	10,000	0.69	64.11
Truck	92	26,000	2,392,000	5 00	3.89	1860.98
Total						2343.69

GEV Gross energy value

strength—expressed as leaf surface—to estimate the mass available and then the time necessary to trim. Apparently, pruner's choice plays a key rule in the quantity of biomass produced since the bigger trees present in site C provided only slightly more matter than the ones in site B.

Productivity in extraction is quite low compared to other biomass recovery operations as for example the thinning of pine plantations operated by a tractor with a grapple. In fact, although the productivity recorded by Spinelli and Nati (2009) was higher (2.2 t_{dw} PMH_{15}^{-1}), the results presented by Savelli et al. (2010) carried out in a coniferous stand were quite close to those of the present study, recording a productivity between 0.98 t_{dw} PMH_{15}^{-1} and 1.28 t_{dw} PMH_{15}^{-1}. Nevertheless, the comparison is hard to stand because whole trees aligned in piles is a very different material from crooked branches, fallen untidily on the ground. The Authors think that the deployment of a mini-excavator mounting a grapple or a pitchfork, it could be a valuable option to speed up the collection of branches, in stands with a slope up to 15%, as the case of site A in the present study. Rubber-tired mini-skidders can effectively replace forestry-adapted farm tractors under certain conditions, as outlined in the previous studies (Savelli et al. 2010; Spinelli et al. 2012).

Processing affected only site A and is very difficult to compare the productivity carried out in this study to other examples in the literature due to the peculiarity of the context. Normally, the processing phase, when performed not in a high mechanized way, affects the bucking of logs (not branches) or is included in the felling phase, as for example in Picchio et al. (2009).

Transportation covers in both cases just a brief distance, from the secondary road to the comminution centre. The little number of loads did not allow to model this phase and simulate the best option to choose, that is the transport of un-comminuted material or the chipping at the road, followed by a later transportation to the biomass plant.

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Table 7 Energy inputs needed for each operative step in the wood biomass recovery

Site A			
DIRECT energy inputs			
Pruning (GJ t_{dw}^{-1})	0.05	Pruning (GJ ha^{-1})	1.40
Extraction (GJ t_{dw}^{-1})	0.12	Extraction (GJ ha^{-1})	3.52
Transportation (GJ t_{dw}^{-1})	0.02	Transportation (GJ ha^{-1})	0.64
Chipping (GJ t_{dw}^{-1})	0.15	Chipping (GJ ha^{-1})	4.33
Total	0.34		9.89
INDIRECT energy inputs			
Pruning (GJ t_{dw}^{-1})	0.00	Pruning (GJ ha^{-1})	0.00
Extraction (GJ t_{dw}^{-1})	0.06	Extraction (GJ ha^{-1})	1.80
Transportation (GJ t_{dw}^{-1})	0.19	Transportation (GJ ha^{-1})	5.54
Chipping (GJ t_{dw}^{-1})	0.02	Chipping (GJ ha^{-1})	0.59
Total	0.27		7.92
Site B			
DIRECT energy inputs			
Cleaning (GJ t_{dw}^{-1})	0.00	Cleaning (GJ ha^{-1})	0.04
Pruning (GJ t_{dw}^{-1})	0.05	Pruning (GJ ha^{-1})	1.23
Extraction (GJ t_{dw}^{-1})	0.22	Extraction (GJ ha^{-1})	5.62
Transportation (GJ t_{dw}^{-1})	0.03	Transportation (GJ ha^{-1})	0.75
Chipping (GJ t_{dw}^{-1})	0.23	Chipping (GJ ha^{-1})	5.99
Total	0.53		13.63
INDIRECT energy inputs			
Cleaning (GJ t_{dw}^{-1})	0.00	Cleaning (GJ ha^{-1})	0.01
Pruning (GJ t_{dw}^{-1})	0.00	Pruning (GJ ha^{-1})	0.00
Extraction (GJ t_{dw}^{-1})	0.04	Extraction (GJ ha^{-1})	1.16
Transportation (GJ t_{dw}^{-1})	0.30	Transportation (GJ ha^{-1})	7.68
Chipping (GJ t_{dw}^{-1})	0.03	Chipping (GJ ha^{-1})	0.81
Total	0.38		9.67

The hourly productivity of chipping was in line with the comminution of the tops of other species like poplar in the study of Assirelli et al. (2013), where a productivity of 11 t_{dw} PMH_0^{-1} is reported, or the chipping of logging residues in Asikainen and Pulkkinen (1998) with a productivity of 11.5 t_{dw} PMH_0^{-1}, material similar to chestnut branches.

A natural drying of four weeks or more generally appears to be a promising method for increasing the energy content of woody feedstocks (Cutshall et al. 2013). The recommendation rose from the present study is that a seasoning of four or more months allows a moisture loss up to 30% of m. c.

Fines (chips smaller than 3.15 mm) are more frequent in site A than in site B, but while they are under the percentage fraction required by the class, oversize particles, present in both sites prevent from achieving the target of fitting the P16B class. The production of a high quantity of coarse wood particles seems to be in line with the results obtained by Nati et al. (2010) with hybrid poplar, where the tendency to produce oversize particles was put in relation not only with wood tensile strength but even with the presence of flexible twigs in the material to be chipped. In Nati et al. (2014), the same chipping machine gave significantly higher amounts of oversized material when fed with logging residues, resulting in a product unsuitable for non-industrial biomass systems. In Spinelli et al. (2011b), branches of softwood species showed a tendency to produce a large proportion of large particles, even within the limits of acceptable chip sizes, when compared to stem material.

The pruning and biomass recovery operation in chestnut orchards costed 112.9 € t_{dw}^{-1} in site A and

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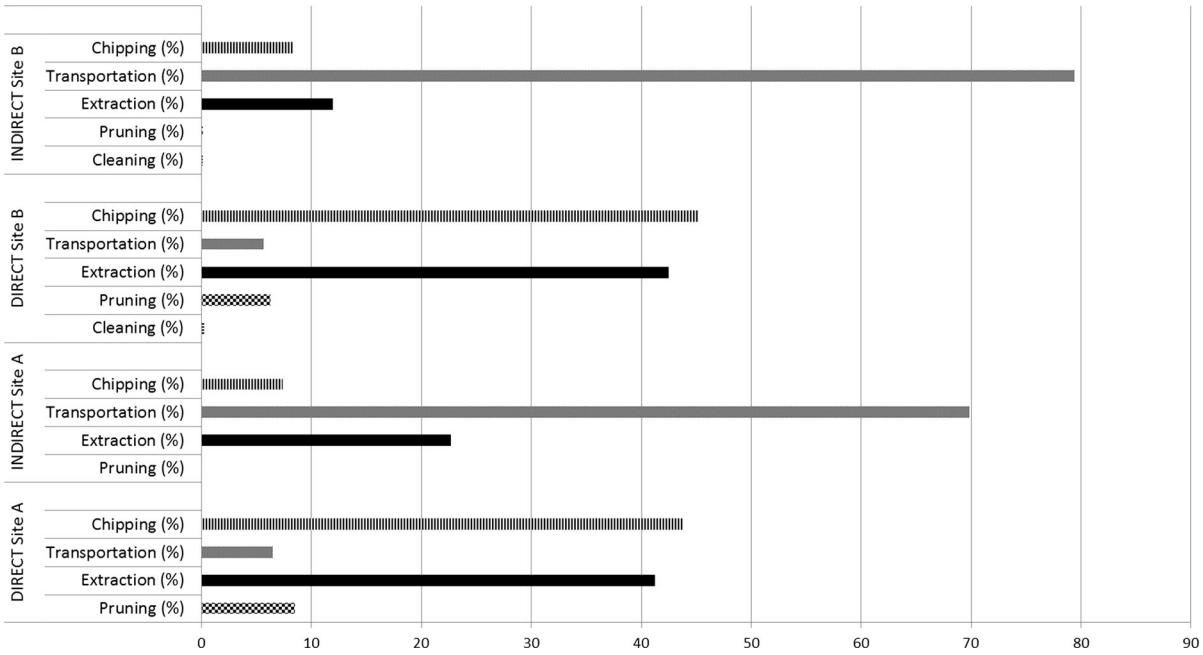


Fig. 6 Allocation of the energy requested for recovery phase—on percentage (%)—for the two sites

188.6 € t_{dw}^{-1} in site B. It is not an easy task to find analogous situations of comparison with the chestnut pruning. For example, the production of biomass from the removing (not pruning) of ancillary species in a hardwood plantation costed from to 50.4 or 52.4 € t_{dw}^{-1} (Magagnotti et al. 2011) including felling, extraction and chipping using a low mechanization level, as in the present study. If one in the chestnut case subtracts the transportation of un-comminuted material, avoidable by chipping on the road, the cost decreases only to 83.5 € t_{dw}^{-1} for site A and to 129.2 € t_{dw}^{-1} for site B. The costs per hectare reached for chestnut are by far higher compared—for example—to those achieved in coniferous stands as described in Beinhofer (2010) who reports a cost for oak and beech stands from 150 € ha^{-1} to 347 € ha^{-1} depending on tree species and age; or Erikkson (2004), who calculated for scots pine a cost from 313 € ha^{-1} up to 1078 € ha^{-1} depending on trees' density. It is worth noting that an extraction system based on tractor and a winch resulted only +3% expensive than tractor and a trailer, encouraging in this way the use of this kind of equipment on slope terrain, even for little amount of biomass as in this case.

The direct and indirect consumption of fossil energy is just one aspect of the environmental impact of the activities affecting biomass recovery, and its use as the main evaluation parameter may appear to be restrictive. Never the less, the deployment of fossil energy remains a good indicator of a system sustainability (Magagnotti and Spinelli 2012; Vusić et al. 2013; Maesano et al. 2013). The favourable input/output ratio achieved in the study confirms the basic requirement for any bioenergy generation system, that is, the energy produced must be greater than the inputs of non-renewable energy required to establish and operate the scheme. (Matthews 2001). Site A emerged as the most energetic efficient system, with an input/output ratio of 30:1, in line with other systems, as the pine thinning operation reported by Baldini et al. (2010).

Certainly, terrain relief expressed as accessibility (e.g. slope) can affect not only stand quality of sweet chestnut (Díaz-Varela et al. 2011), performing as socioeconomic variable related to the management intensity of the stands (Zerger et al. 2009), but even conditioning the operating choices during pruning.

Many variables that are kept into account for operating an efficient and sustainable biomass recovery in chestnut orchards and the arguments that arise, although they do not allow a detailed comparative analysis, can be a good starting point for further research.

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Compliance with ethical standards

Conflict of interest None.

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